cfg_we

cfg_addr[3:0] enable_relu

cfg_wdata[7:0]

rst_n
clk
s_valid

s_data[7:0] win_reg[2][0][7:0] RTL_REG_ASYNC C

CECLR D Q win_reg[2][1][7:0] RTL_REG_ASYNC C
CECLR D Q win_reg[2][2][7:0] RTL_REG_ASYNC C

CECLR D Q
lb0_reg[0][7:0] RTL_REG C

CED Q

lb0_reg[1][7:0] RTL_REG C

CED Q

lb0_reg[2][7:0] RTL_REG C

CED Q

lb0_reg[3][7:0] RTL_REG C

CED Q

lb0_reg[4][7:0] RTL_REG C

CED Q

lb0_reg[5][7:0] RTL_REG C

CED Q

lb0_reg[6][7:0] RTL_REG C

CED Q

lb0_reg[7][7:0] RTL_REG C

CED Q

lb0_reg[8][7:0] RTL_REG C

CED Q

lb0_reg[9][7:0] RTL_REG C

CED Q

lb0_reg[10][7:0] RTL_REG C

CED Q

lb0_reg[11][7:0] RTL_REG C

CED Q

lb0_reg[12][7:0] RTL_REG C

CED Q

lb0_reg[13][7:0] RTL_REG C

CED Q

lb0_reg[14][7:0] RTL_REG C

CED Q

lb0_reg[15][7:0] RTL_REG C

CED Q

lb0_reg[16][7:0] RTL_REG C

CED Q

lb0_reg[17][7:0] RTL_REG C

CED Q

lb0_reg[18][7:0] RTL_REG C

CED Q

lb0_reg[19][7:0] RTL_REG C

CED Q

lb0_reg[20][7:0] RTL_REG C

CED Q

lb0_reg[21][7:0] RTL_REG C

CED Q

lb0_reg[22][7:0] RTL_REG C

CED Q

lb0_reg[23][7:0] RTL_REG C

CED Q

lb0_reg[24][7:0] RTL_REG C

CED Q

lb0_reg[25][7:0] RTL_REG C

CED Q

lb0_reg[26][7:0] RTL_REG C

CED Q

lb0_reg[27][7:0] RTL_REG C

CED Q

lb0_reg[28][7:0] RTL_REG C

CED Q win_reg[1][0][7:0] RTL_REG_ASYNC C

CECLR D Q win_reg[1][1][7:0] RTL_REG_ASYNC C

CECLR D Q win_reg[1][2][7:0] RTL_REG_ASYNC C

CECLR D Q

lb1_reg[0][7:0] RTL_REG C

CED Q

lb1_reg[1][7:0] RTL_REG C

CED Q

lb1_reg[2][7:0] RTL_REG C

CED Q

lb1_reg[3][7:0] RTL_REG C

CED Q

lb1_reg[4][7:0] RTL_REG C

CED Q

lb1_reg[5][7:0] RTL_REG C

CED Q

lb1_reg[6][7:0] RTL_REG C

CED Q

lb1_reg[7][7:0] RTL_REG C

CED Q

lb1_reg[8][7:0] RTL_REG C

CED Q

lb1_reg[9][7:0] RTL_REG C

CED Q

lb1_reg[10][7:0] RTL_REG C

CED Q

lb1_reg[11][7:0] RTL_REG C

CED Q

lb1_reg[12][7:0] RTL_REG C

CED Q

lb1_reg[13][7:0] RTL_REG C

CED Q

lb1_reg[14][7:0] RTL_REG C

CED Q

lb1_reg[15][7:0] RTL_REG C

CED Q

lb1_reg[16][7:0] RTL_REG C

CED Q

lb1_reg[17][7:0] RTL_REG C

CED Q

lb1_reg[18][7:0] RTL_REG C

CED Q

lb1_reg[19][7:0] RTL_REG C

CED Q

lb1_reg[20][7:0] RTL_REG C

CED Q

lb1_reg[21][7:0] RTL_REG C

CED Q

lb1_reg[22][7:0] RTL_REG C

CED Q

lb1_reg[23][7:0] RTL_REG C

CED Q

lb1_reg[24][7:0] RTL_REG C

CED Q

lb1_reg[25][7:0] RTL_REG C

CED Q

lb1_reg[26][7:0] RTL_REG C

CED Q %

RTL_MOD

RTL_MOD

I0[3:0]

I1[1:0] O[1:0]

lb1_reg[27][7:0] RTL_REG C

CED Q

kernel_reg[0][0]_i RTL_ROM

A[31:0] O

kernel_reg[0][1]_i RTL_ROM

A[31:0] O /RTL_DIV RTL_DIV

I0[3:0]

I1[1:0] O[2:0] <RTL_LT RTL_LT O
I0[3:0] V=B"1001" I1[3:0]kernel_reg[0][2]_i RTL_ROM

A[31:0] O

lb1_reg[28][7:0] RTL_REG C

CED Q

kernel_reg[1][0]_i RTL_ROM

A[31:0] O

kernel_reg[1][1]_i RTL_ROM

A[31:0] O

kernel_reg[1][2]_i RTL_ROM

A[31:0] O

kernel_reg[2][0]_i RTL_ROM

A[31:0] O

kernel_reg[2][1]_i RTL_ROM

A[31:0] O

kernel_reg[0][0]_i__0 RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0] kernel_reg[0][1]_i__0 RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0] RTL_AND
RTL_AND

I0

I1 O

kernel_reg[0][2]_i__0 RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0] win_reg[0][0][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[1][0]_i__0 RTL_MUX

S=32'b00000000000000000000000000000001 I0

S=default I1 O

S[31:0] kernel_reg[1][1]_i__0 RTL_MUX

S=32'b00000000000000000000000000000001 I0

S=default I1 O

S[31:0] kernel_reg[1][2]_i__0 RTL_MUX

S=32'b00000000000000000000000000000001 I0

S=default I1 O

S[31:0] kernel_reg[2][0]_i__0 RTL_MUX

S=32'b00000000000000000000000000000010 I0

S=default I1 O

S[31:0] kernel_reg[2][1]_i__0 RTL_MUX

S=32'b00000000000000000000000000000010 I0

S=default I1 O

S[31:0]

kernel_i RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__0 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S win_reg[0][1][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_i__2 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__3 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__4 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__5 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_i__6 RTL_MUX

S=1'b1 I0

S=default I1 O S

kernel_reg[0][0][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[0][1][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[2][2]_i RTL_ROM

A[31:0] O win_reg[0][2][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[0][2][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[1][0][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[1][1][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[1][2][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[2][0][7:0] RTL_REG_ASYNC C

CECLR D Q

kernel_reg[2][1][7:0] RTL_REG_ASYNC C

CECLR D Q *RTL_MULT RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__0 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] kernel_reg[2][2]_i__0 RTL_MUX

S=32'b00000000000000000000000000000010 I0

S=default I1 O

S[31:0]+

col_cnt0_i RTL_ADD

I1

I0[4:0] O[4:0] *RTL_MULT__1 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__2 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__3 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__4 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__5 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] *RTL_MULT__6 RTL_MULT

I0[8:0]

I1[7:0] O[15:0]

prod_reg[0][0][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[0][1][15:0] RTL_REG_ASYNC C

CLR D Q

kernel_i__7 RTL_MUX

S=1'b1 I0

S=default I1 O S

col_cnt_i RTL_MUX

S=5'b11111 I0[4:0] S=default I1[4:0] O[4:0] S[4:0] prod_reg[0][2][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[1][0][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[1][1][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[1][2][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[2][0][15:0] RTL_REG_ASYNC C

CLR D Q

prod_reg[2][1][15:0] RTL_REG_ASYNC C

CLR D Q +

sat_val5_i RTL_ADD

I0[16:0]

I1[16:0] O[16:0] kernel_reg[2][2][7:0] RTL_REG_ASYNC C

CECLR D Q

col_cnt_reg[4:0] RTL_REG_ASYNC C

CECLR D Q +

sat_val5_i__0 RTL_ADD

I0[16:0]

I1[16:0] O[16:0] +

sat_val5_i__1 RTL_ADD

I0[16:0]

I1[16:0] O[16:0] +

sat_val5_i__2 RTL_ADD

I0[16:0]

I1[16:0] O[16:0] +

row_cnt0_i RTL_ADD

I1

I0[4:0] O[4:0]

row_cnt_i__0 RTL_ROM

A[4:0] O *RTL_MULT__7 RTL_MULT

I0[8:0]

I1[7:0] O[15:0] +

sat_val4_i RTL_ADD

I0[17:0]

I1[17:0] O[17:0] +

sat_val4_i__0 RTL_ADD

I0[17:0]

I1[17:0] O[17:0]

row_cnt_i RTL_MUX

S=5'b11111 I0[4:0] S=default I1[4:0] O[4:0] S[4:0]

row_cnt_i__1 RTL_MUX

S=1'b1 I0

S=default I1 O S prod_reg[2][2][15:0] RTL_REG_ASYNC C

CLR D Q +

sat_val3_i RTL_ADD

I0[18:0]

I1[18:0] O[18:0]

row_cnt_reg[4:0] RTL_REG_ASYNC C

CECLR D Q +

sat_val2_i RTL_ADD

I0[19:0]

I1[19:0] O[19:0] >=

raw_win_valid1_i RTL_GEQ O
I0[4:0] V=B"00010" I1[4:0] <

sat_val1_i RTL_LT O

I0[19:0] V=X"F8000" I1[19:0]

>=

raw_win_valid0_i__0 RTL_GEQ O

I0[4:0] V=B"00010" I1[4:0] raw_win_valid0_i RTL_AND

I0

I1 O

sat_val_i RTL_MUX S
V=B"1000000000000000", S=1'b1 I0[15:0] S=default I1[15:0] O[15:0] >

sat_val1_i__0 RTL_GT O

I0[19:0] V=X"07FFF" I1[19:0]

raw_win_valid_i RTL_AND

I0

I1 O

sat_val_i__0 RTL_MUX S

V=B"0111111111111111", S=1'b1 I0[15:0] S=default I1[15:0] O[15:0]

win_valid_reg RTL_REG_ASYNC C

CLR D Q

post_proc_val1_i RTL_AND

I0

I1 O

val_stg1_reg RTL_REG_ASYNC C

CLR D Q

post_proc_val_i RTL_MUX S

S=1'b1 I0[15:0] S=default I1[15:0] O[15:0]

m_valid_reg RTL_REG_ASYNC C

CLR D Q m_data_reg[15:0] RTL_REG_ASYNC C

CLR D Q m_valid m_data[15:0]

2:0

2:0

2:0

2:0

2:0

2:0

2:0

2:0

2:0

1:0

1:0

1:0

1:0

1:0

1:0

1:0

1:0

1:0

...

7:0

7:0

7:0

15

7:0

7:0

7:0

7:0

7:0

7:0